Hospital Data Analysis Report

Massachusetts General Hospital — Power Pivot Analytics

Analysis of ~1,000 synthetic patient records from 2011–2022,
covering demographics, insurance coverage, and medical encounters & procedures.

Data Source: Maven Analytics Data Playground & Synthea™ Synthetic Patient Generator

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1. Objective

The objective of this report is to build a high-level KPI dashboard for the executive team based on a subset of patient records. The report provides stakeholders with visibility into the hospital's recent performance across encounter volumes, cost and coverage, procedural frequency, and patient behavior trends. This analysis leverages approximately 1,000 synthetic patient records spanning 2011–2022 to surface actionable insights and establish a foundation for ongoing performance monitoring.

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2. Data Dictionary

The dataset comprises synthetic patient records from Massachusetts General Hospital spanning 2011–2022. Below is the data dictionary for key fields used in the analysis.

Field Name	Description
Patient ID	Unique identifier for each patient
Encounter ID	Unique identifier for each medical encounter
Encounter Class	Category of encounter (Ambulatory, Outpatient, Wellness, Urgent Care, Emergency, Inpatient)
Encounter Date	Date and time of the encounter
Encounter Duration	Length of the encounter
Procedure Code	Standardized code for the medical procedure performed
Procedure Description	Human-readable description of the procedure

Field Name	Description
Base Cost	Standard cost associated with the procedure
Total Claim Cost	Total amount claimed for the encounter
Payer	Insurance provider or payer entity
Payer Coverage	Amount covered by the payer/insurance
Patient Demographics	Age, gender, race, ethnicity, and geographic information
Readmission Flag	Indicator for whether the patient was readmitted within 30 days

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3. Encounters Overview

3.1 Total Encounters by Year

The table below presents the total number of patient encounters recorded at MGH for each year in the analysis period. Tracking annual encounter volume is a foundational metric for understanding hospital utilization and capacity trends.

Year	Total Encounters
2011	1,336
2012	2,107
2013	2,495
2014	**3,887**
2015	2,468
2016	2,451
2017	2,360
2018	2,290
2019	2,230
2020	2,519
2021	**3,531**
Total	**27,674**

Key Trends

2014 recorded the highest encounter volume at **3,887**, representing a significant ramp-up from the initial 1,336 encounters in 2011. Following this peak, volumes stabilized between **2,230 and 2,519** encounters annually from 2015 through 2020. A notable resurgence occurred in **2021**, with encounters climbing to **3,531** — an increase of **40.2%** over 2020. The total across all years sums to **27,674 encounters**. Note: Records from 2022 are excluded because they only cover a period of 2 months.

3.2 Encounter Class Distribution

Each encounter at MGH is classified into one of six encounter classes, enabling the executive team to monitor shifts in care delivery patterns over time. The classes tracked in this analysis are:

- **Ambulatory** — Scheduled outpatient visits where patients are seen and discharged the same day, including routine check-ups and follow-up appointments.
- **Outpatient** — Hospital-based services that do not require overnight admission, such as diagnostic testing and minor procedures.
- **Wellness** — Preventive care encounters including annual physical, screenings, and immunizations.
- **Urgent Care** — Encounters for conditions requiring prompt attention but not severe enough to warrant emergency department visits.
- **Emergency** — Unscheduled visits to the emergency department for acute conditions and injuries.
- **Inpatient** — Admissions require at least one overnight stay, typically involving complex diagnoses, surgeries, or intensive monitoring.

For each year in the analysis period, the percentage breakdown across these encounter classes was examined to identify meaningful shifts in care delivery. Monitoring class distribution enables MGH to align staffing, resource allocation, and facility planning with evolving patient demand patterns.

3.3 Encounter Duration

Encounter duration is a critical operational metric that reflects patient throughput, resource utilization, and acuity levels. The table below summarizes the distribution of encounters by duration category.

Duration Category	Percentage
Under 24 Hours	**95.90%**
Above 24 Hours	4.10%

The overwhelming majority of encounters (**95.9%**) were resolved within 24 hours, indicating that MGH primarily handles short-duration visits — consistent with a high-volume ambulatory and outpatient care model. Only **4.1%** of encounters extended beyond 24 hours, typically representing inpatient stays and complex surgical or interventional procedures. This distribution underscores the importance of optimizing same-day care workflows while maintaining adequate capacity for extended-stay patients.

3.4 Procedures Covered by Insurance

Across the full analysis period, a total of **24,791 procedures** were covered by insurance. This figure represents the volume of procedures for which MGH received payer reimbursement, serving as a key indicator of revenue cycle health. High insurance coverage rates reduce bad debt exposure and support predictable cash flow. Monitoring this metric over time allows the finance team to detect trends in payer mix, denial rates, and prior authorization outcomes — all of which directly impact the hospital's bottom line and operational sustainability.

3.5 Average Length of Stay

Understanding the average length of stay (ALOS) is a critical operational metric for MGH. It directly influences bed availability, staffing requirements, and overall hospital throughput. ALOS serves as a benchmark for evaluating care efficiency. A shorter average stay may indicate streamlined care pathways and effective discharge planning, while a longer stay could reflect complex case mixes, complications, or delays in post-acute placement. Monitoring ALOS over time and across encounter classes allows the executive team to identify bottlenecks, set performance targets, and align capacity planning with actual utilization patterns.

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4. Cost & Coverage Insights

4.1 Average Cost per Visit

Tracking the average cost per visit is one of the most important financial KPIs for hospital leadership. This metric provides a consolidated view of operational spending intensity per patient encounter, encompassing direct costs (clinical labor, supplies, medications) and indirect costs (facilities, administrative overhead). By monitoring average cost per visit on a

quarterly and annual basis, the executive team can benchmark MGH's performance against regional and national peer institutions, identify periods of cost escalation, and pinpoint cost optimization opportunities. Trends in this metric also serve as early indicators of changes in patient acuity, procedural complexity, or operational inefficiency.

4.2 Zero Payer Coverage Encounters

Encounters with zero payer coverage represent cases where no insurance reimbursement was received by the hospital. These encounters are significant for several reasons: they may indicate **uninsured patients** who lack any form of coverage, **denied claims** resulting from coding errors or prior authorization failures, or **self-pay scenarios** where patients elect to pay out-of-pocket. This metric is critical for financial planning, as zero-coverage encounters directly affect the hospital's revenue realization rate and operating margin. Additionally, a high volume of zero-coverage encounters may signal the need for expanded financial counseling services, improved claims processing workflows, or increased charity care provisions. Tracking this metric supports MGH's compliance with community benefit reporting requirements and informs strategic decisions regarding payer contract negotiations.

4.3 Top 10 Most Frequent Procedures

The analysis identified the **top 10 most frequently performed procedures** across the study period, along with their associated average base costs. Understanding procedural volume drivers is essential for capacity planning, supply chain management, and revenue forecasting. High-frequency procedures — such as routine assessments, immunizations, and standard diagnostic tests — typically generate significant aggregate revenue despite relatively lower per-unit costs. By pairing frequency data with average base cost, the executive team gains insight into which procedures drive the greatest overall financial volume and where economies of scale may be achievable through standardized care pathways and bulk purchasing agreements.

4.4 Top 10 Highest-Cost Procedures

Separately from volume-driven procedures, the analysis ranked the **top 10 procedures by average base cost** alongside their frequency counts. This view reveals high-value procedures that, while potentially less frequent, significantly impact the hospital's revenue and resource allocation. Procedures such as complex surgeries, advanced imaging, and specialized interventions typically appear in this category. Understanding the cost-frequency relationship for these procedures is vital for capital budgeting (equipment, OR time), physician workforce planning, and payer contract optimization. Procedures with high cost but low volume may also represent growth opportunities if supported by expanded referral networks or center-of-excellence designations.

4.5 Average Total Claim Cost by Payer

The average total claim cost was analyzed by payer (insurance provider) to evaluate payer mix profitability. This breakdown enables MGH to compare reimbursement levels across commercial insurers, Medicare, Medicaid, and other payer categories. Significant variation in average claim cost by payer often reflects differences in patient acuity within each payer segment, contracted reimbursement rates, and coverage scope. These insights are directly actionable: they support data-driven payer contract negotiations, inform managed care strategy, and help the finance team model the revenue impact of shifts in payer mix such as an increasing share of Medicare Advantage enrollees or Medicaid expansion populations.

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5. Patient Behavior Analysis

5.1 Unique Patient Admissions by Quarter

The number of unique patients admitted to MGH was tracked on a **quarterly basis** across the full 2011–2022 analysis period. Quarterly granularity provides several advantages over annual aggregates: it enables identification of **seasonal patterns** (e.g., flu season surges, summer injury spikes), reveals underlying **growth or contraction trends** in the patient population, and supports more precise **capacity planning** ensuring that staffing, bed availability, and supply chain resources align with anticipated demand.

Monitoring unique patient counts — as distinct from total encounters — eliminates double-counting of patients with multiple visits within the same quarter. This provides a clearer picture of the hospital's reach and market penetration. Notable deviations from expected seasonal patterns (such as an unusual dip or spike in each quarter) can trigger deeper investigation into contributing factors, including community health events, competitor dynamics, or changes in referral patterns.

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6. Future Scope

Building on the foundational analysis presented in this report, the following analytics enhancements are planned to deepen MGH's operational and clinical intelligence:

Initiative	Description
30-Day Readmission Analysis	Identifying patients readmitted within 30 days of a previous encounter. This is a critical quality metric tied to CMS penalties under the Hospital Readmissions Reduction Program (HRRP) and is directly correlated with patient outcomes and care coordination effectiveness.
High-Readmission Patient Profiling	Determining which patients had the most readmissions across the study period to enable targeted care coordination, transitional care programs, and individualized intervention plans aimed at reducing preventable readmissions.
Predictive Analytics	Developing machine learning models to predict readmission risk, expected length of stay, and cost forecasting at the patient level. These models would enable proactive resource allocation and early clinical intervention.
Population Health Insights	Conducting deeper demographic and geographic analysis of the patient population to identify health disparities, chronic disease prevalence, and underserved communities — supporting MGH's community health improvement initiatives.